Cross-Currency Dynamics in Cryptocurrencies

Abstract

When Silicon Valley Bank collapsed in March 2023, USDC briefly traded at \$0.87 on some venues (Kraken trough: \$0.874), fragmenting Bitcoin prices across exchanges and quote currencies. Using 1-minute data from Binance, Coinbase, and Kraken, we decompose this fragmentation into a large stablecoin-driven component—USDC crisis-mean unadjusted dispersion near $+320$ basis points—and a smaller adjusted residual around $+5$ bps after marking the stablecoin leg to USD, with mean-reversion half-lives of 0.54–1.10 minutes across robustness specifications. The decomposition reveals a two-layer dynamic: the adjusted basis B_t mean-reverts in ≈ 0.6 minutes during the crisis, while the underlying USDC peg deviation half-life extends to ≈ 572 minutes—a $\sim 940\times$ gap showing that persistent dislocation resides in the peg layer, not in cross-market arbitrage efficiency. A Chow structural-break test at SVB onset rejects parameter stability for both dispersion levels ($F = 2{,}089$) and AR(1) dynamics ($F = 71$, both $p < 10^{-16}$), and USDC crisis adjusted-residual excess kurtosis reaches 12.9 versus 3.8 pre-SVB (P99 expands from 13.6 to 81.4 bps, a $6\times$ tail blowout), confirming USDC-specific tail risk. Simultaneously, USDT traded at a $+58$ bps premium as traders rotated away from USDC, and the cross-stablecoin adjusted-residual correlation dropped from 0.44 to 0.16, consistent with divergent stablecoin dynamics rather than a common shock. Fiat-quoted BTC leads price discovery over USDT-quoted BTC on Kraken (Hasbrouck IS midpoint 0.73), while the USDC channel does not establish a stable cointegrating relationship. Intra-exchange USDC crisis arbitrage profitability falls from 29% (fee-only) to 7% (fee+slippage)—versus 0% pre-SVB under fee+slippage—and cross-exchange BTC/USD crisis profitability falls to about 2.3% under fee+slippage bounds (BTC/USDT cross-exchange: ≈ 1.8%, denominated in USDT). These findings support a directional policy implication: the GENIUS Act's reserve-composition, timely-redemption, transparency, and bankruptcy super-priority provisions target the failure chain highlighted by the SVB episode, and an illustrative counterfactual maps 25%–75% mitigation of reserve lock-up to implied crisis D_t of 240–80 bps versus the observed 320 bps baseline.

1 Context and Motivation

Stablecoins are the primary quote currency on most crypto exchanges, functioning as digital cash accounts that traders use to quote prices, move liquidity, and fund positions across venues. Because stablecoins are not identical to bank deposits—redeemability, custody, and transparency can differ—their prices can deviate from \$1 and introduce a *stablecoin funding risk premium*: the same underlying asset (BTC) then trades at different nominal prices across quote currencies (BTC/USD vs. BTC/USDC vs. BTC/USDT) purely because of stablecoin-specific risk. In 2025, the U.S. enacted the GENIUS Act, establishing a federal framework for dollar-pegged stablecoins with stricter reserve backing and transparency, targeting this channel [11].

To study how stablecoin stress affects pricing, we analyze March 1–21, 2023 (UTC), which includes the SVB failure. When SVB collapsed on March 10, approximately \$3.3B of Circle's USDC reserves became temporarily inaccessible, and USDC de-pegged to as low as \$0.87 on some venues (\$0.874 on Kraken) on March 11–12 [3]. This episode provides a clean, high-frequency stress test of cross-currency basis behavior, liquidity fragmentation, and price discovery—and a natural case study for what the GENIUS Act is designed to prevent.

2 Data and Methodology

2.1 Data sources and preprocessing

We collect 1-minute OHLCV candles for BTC quoted in USD, USDT, and USDC from three centralized exchanges chosen for their distinct regulatory exposure and market structure. Binance provides BTC/USDT and BTC/USDC and dominates global USDT volume. Coinbase provides BTC/USD, BTC/USDT, and USDT/USD; it is a primary U.S. fiat gateway. Kraken provides BTC/USD, BTC/USDT, BTC/USDC, USDT/USD, and USDC/USD, offering the broadest set of stablecoin-to-fiat legs. BTC/EUR pairs are collected for completeness but are not central to the results. For stablecoin-to-USD conversion in B_t, we use

venue-specific legs (e.g., Kraken USDC/USD for Kraken channels). All series are aligned to a unified 1-minute UTC grid. Overall coverage is high for most series, but it is uneven by regime: pre-SVB Kraken BTC/USDC coverage falls to 89.6% while crisis/post coverage is near-complete. Short micro-gaps (up to 5 minutes) are forward-filled; larger gaps remain as missing values (NaN).

Table 1: Core series data coverage and forward-fill exposure. Coverage is the share of non-missing 1-minute observations on the unified UTC grid; forward-fill share is the percent of minutes filled by carry-forward (up to 5 minutes).

Pair	Coverage Overall (%)	Coverage Pre-SVB (%)	Coverage Crisis (%)	Coverage Post-SVB (%)	Forward-Fill Share Overall (%)
Kraken BTC/USD	99.98	100.00	100.00	99.95	0.18
Kraken BTC/USDT	98.44	97.36	99.81	99.06	30.84
Kraken BTC/USDC	94.05	89.62	99.79	96.57	42.07
Kraken USDC/USD	97.80	96.03	99.35	99.06	28.57
Kraken USDT/USD	99.98	100.00	100.00	99.95	0.91
Coinbase BTC/USD	99.10	97.90	100.00	100.00	0.06
Coinbase BTC/USDT	98.93	97.53	99.95	99.98	17.98
Coinbase USDT/USD	99.02	97.72	100.00	100.00	0.06
Binance BTC/USDT	100.00	100.00	100.00	100.00	0.00
Binance BTC/USDC	100.00	100.00	100.00	100.00	0.00

Because lower pre-SVB USDC coverage can bias calm-regime baselines, we treat pre-SVB USDC level comparisons cautiously and rely on no-forward-fill robustness checks for mean-reversion and cointegration specifications.

Table 2: Forward-fill sensitivity for selected headline metrics. All-sample uses the baseline aligned 1-minute series; No-FF sample removes observations with forward-filled inputs for each metric. Signs are largely stable, but magnitudes can move in thinner channels and rolling-volatility windows can become sparse.

Metric	Statistic	All	No-FF	Δ (No-FF-All)	N	N (No-FF)	Retention (%)
USDC D_t (Kraken, crisis)	Mean bps	319.974	360.511	40.537	4311	3324	77.100
USDC B_t (Kraken, crisis)	Mean bps	5.271	6.370	1.099	4283	3002	70.100
USDT premium (Kraken, crisis)	Mean bps	58.114	58.128	0.014	4320	4319	100.000
USDT premium (Coinbase, crisis)	Mean bps	59.680	59.721	0.041	4320	4317	99.900
BTC/USDT basis (BN-KR, calm)	Mean bps	0.157	0.061	-0.096	25456	17098	67.200
USDC arb (Kraken, 3-leg, crisis)	Fee-only %	29.162	35.143	5.981	4283	3002	70.100
USDC arb (Kraken, 3-leg, crisis)	Fee+slip %	6.724	7.295	0.571	4283	3002	70.100
RV 60m (KR BTC/USDC, crisis)	Mean bps/hr	136.991	412.777	275.786	4105	431	10.500
RV 60m (KR BTC/USDT, crisis)	Mean bps/hr	66.623	99.348	32.725	4009	142	3.500

Table 2 quantifies all-sample vs. strict no-forward-fill sensitivity for headline metrics. Core signs are stable (e.g., crisis USDC $D_t > 0$, USDT premium > 0, and fee+slippage profitability below fee-only), but some USDC magnitudes shift on thinner no-FF samples, and rolling-volatility windows become sparse under strict no-FF filters. We therefore interpret high-frequency level magnitudes as directional unless no-FF retention is adequate.

We proxy transaction frictions using the intra-candle range:

$$\text{Intra-minute Range (bps)} = \left(\frac{\text{High}_t - \text{Low}_t}{\text{Close}_t}\right) \times 10000. \tag{1}$$

This is a coarse range proxy that mechanically loads on volatility and microstructure noise; we interpret it cautiously as a friction indicator, not a quoted bid–ask spread.

2.2 Core quantitative objects

For stablecoin $Q \in \{\text{USDC}, \text{USDT}\}$, we track two objects. The *unadjusted dispersion*:

$$D_{Q,t} = \big(\ln(P_{\text{BTC}/Q,t}) - \ln(P_{\text{BTC/USD},t})\big) \times 10000, \tag{2}$$

and the *adjusted parity residual*:

$$B_{Q,t} = \big(\ln(P_{\text{BTC}/Q,t} \times P_{Q/\text{USD},t}) - \ln(P_{\text{BTC/USD},t})\big) \times 10000. \tag{3}$$

These satisfy $B_{Q,t} = D_{Q,t} + 10000 \cdot \ln(P_{Q/\text{USD},t})$, so large stablecoin de-pegs mechanically drive D_t while B_t captures the residual after marking the stablecoin leg to USD. Both objects are computed intra-exchange to isolate stablecoin effects from venue differences; cross-venue deviations are studied separately as cross-exchange log-price differentials. Unless explicitly noted otherwise, basis and dispersion quantities are reported in log-bps; peg deviations, intra-minute ranges, and realized-volatility summaries are reported in arithmetic bps-derived units.

2.3 Mean reversion via Ornstein–Uhlenbeck

We model B_t as an OU process $dX_t = \theta(\mu - X_t)dt + \sigma dW_t$ and estimate the discrete AR(1) form $X_t = c + \rho X_{t-1} + \varepsilon_t$. Stationarity is checked with ADF tests (with large N, rejection is expected and serves as a baseline screen). The half-life uses the exact mapping:

$$\tau_{1/2} = \frac{\ln(2)\,\Delta t}{-\ln(\rho)}, \quad 0 < \rho < 1. \tag{4}$$

If $\rho \leq 0$ or $\rho \geq 1$, half-life is undefined (NaN). The OU framework assumes Gaussian innovations; crisis-period B_t exhibits substantial excess kurtosis (Section 3.3), so we use the AR(1) model solely for its half-life mapping, not as a full distributional model. We report robustness at 5-minute sampling and with forward-filled minutes removed.

2.4 Price discovery: Johansen + VECM (primary) and Granger (secondary)

For the two primary Kraken channels (BTC/USD vs. BTC/USDC and BTC/USD vs. BTC/USDT), we use synchronized 1-minute log-price levels, dropping forward-filled minutes. We test for cointegration with Johansen trace tests (constant restricted to the cointegration relation; lag order via BIC on the levels VAR) [8, 4]. When rank $= 1$, we estimate:

$$\Delta \mathbf{p}_t = \alpha\beta^\top \mathbf{p}_{t-1} + \sum_{i=1}^{k_\Delta} \Gamma_i \Delta \mathbf{p}_{t-i} + \mathbf{u}_t. \tag{5}$$

The market with smaller $|\alpha|$ adjusts less and is interpreted as the relative leader. We retain BH/FDR-corrected Granger tests on returns as secondary evidence only [6]; with 1-minute bars, asynchrony and large N can overstate economic lead–lag strength.

3 Empirical Results

3.1 Dispersion, adjusted residual, and mean reversion

Before SVB (March 1–9), both D_t and B_t are near zero on Kraken. During the crisis (March 10 00:00 to March 13 00:00 UTC, i.e. three full days), the *unadjusted* USDC dispersion jumps to mean $+319.97$ bps ($\sigma = 322.87$), while the *adjusted* USDC residual is much smaller (mean $+5.27$ bps, $\sigma = 21.21$). USDT shows smaller dislocations in the opposite direction: crisis $D_t = -57.18$ bps, $B_t = +0.68$ bps (Table 3). This asymmetry reflects two simultaneous dynamics: USDC trades as low as \$0.874 on Kraken on March 11 ($\approx -1{,}260$ bps), mechanically inflating D_t, while USDT simultaneously trades at a *premium* to \$1—averaging $+58.1$ bps on Kraken and $+59.7$ bps on Coinbase during the crisis window. The premium pattern is consistent with a within-stablecoin rotation away from USDC under stress, though this interpretation is observational and not a structural causal estimate.

Table 3: Regime Statistics for Unadjusted Dispersion (D_t) and Adjusted Residual (B_t), Kraken

Regime	Series	Mean (bps)	Std (bps)	Mean Abs (bps)	N
Pre-SVB	USDC Kraken D_t (Unadjusted)	0.12	4.95	3.49	11615
Pre-SVB	USDC Kraken B_t (Adjusted)	-0.32	5.00	3.53	11197
Pre-SVB	USDT Kraken D_t (Unadjusted)	-0.54	4.49	3.17	12618
Pre-SVB	USDT Kraken B_t (Adjusted)	-0.60	4.33	3.02	12618
Crisis	USDC Kraken D_t (Unadjusted)	319.97	322.87	321.17	4311
Crisis	USDC Kraken B_t (Adjusted)	5.27	21.21	13.37	4283
Crisis	USDT Kraken D_t (Unadjusted)	-57.18	45.10	58.57	4312
Crisis	USDT Kraken B_t (Adjusted)	0.68	6.57	4.94	4312
Post-SVB	USDC Kraken D_t (Unadjusted)	11.22	21.95	14.27	12514
Post-SVB	USDC Kraken B_t (Adjusted)	0.45	9.06	6.68	12419
Post-SVB	USDT Kraken D_t (Unadjusted)	-30.26	13.29	30.28	12837
Post-SVB	USDT Kraken B_t (Adjusted)	-1.23	6.47	4.92	12837

HAC/Newey–West uncertainty intervals (60-lag cap) remain consistent with the same directional conclusions [9]: USDC crisis D_t is strongly positive (95% CI $[245.7,\ 394.3]$ bps), USDC crisis B_t remains positive (95% CI $[3.6,\ 7.0]$ bps), and crisis USDT premiums are positive on Kraken and Coinbase (95% CIs $[47.4,\ 68.8]$ and $[49.1,\ 70.3]$ bps). Coinbase–Kraken BTC/USD also remains significantly positive (95% CI $[1.7,\ 2.6]$ bps), while Binance–Kraken BTC/USDT calm deviations are economically tiny despite statistical significance.

Table 4: HAC uncertainty for headline mean estimates (Newey–West, 60-lag cap). Values are in basis points with 95% confidence intervals.

Metric	Mean	HAC SE	95% CI low	95% CI high	N
USDC D_t (Kraken, crisis)	319.974	37.908	245.674	394.273	4311
USDC B_t (Kraken, crisis)	5.271	0.873	3.559	6.983	4283
USDT premium (Kraken, crisis)	58.114	5.467	47.399	68.830	4320
USDT premium (Coinbase, crisis)	59.680	5.405	49.087	70.273	4320
BTC/USDT basis (BN-KR, calm)	0.157	0.066	0.028	0.286	25456
BTC/USD basis (CB-KR, full)	2.185	0.235	1.725	2.645	29961

ADF rejects a unit root for all B_t channels ($p < 0.001$). Using the exact half-life mapping, the USDC adjusted residual on Kraken has half-life 0.98 min pre-SVB, 0.61 min in crisis, and 0.79 min post-SVB (1-minute, all observations). Robustness across sampling frequencies (1m, 5m) and forward-fill filters confirms USDC crisis half-life ranges from 0.54 to 1.10 minutes; USDT crisis half-life is longer (0.92–3.06 min). Fast USDC mean reversion is robust, though whether crisis half-life strictly falls relative to calm is specification-sensitive. Crucially, the OU estimation reveals a two-layer dynamic: while the adjusted basis B_t mean-reverts in $\approx$0.6 minutes during crisis ($\hat{\rho} = 0.320$, ADF $p < 0.001$), the underlying USDC peg deviation has a half-life of $\approx$572 minutes ($\hat{\rho} = 0.999$, ADF $p = 0.44$), and USDT peg deviations similarly persist (496–542 minutes on Kraken and Coinbase). This $\sim 940\times$ gap between the peg layer and the arbitrage layer shows that cross-market price correction functions rapidly even in crisis, but the stablecoin-to-USD peg itself—driven by reserve-access risk—is where dislocation persists.

Figure 1: Panel A: stablecoin spot prices vs. USD. Panel B: peg deviations from $1.00 (bps). USDC falls sharply while USDT trades at a premium, consistent with stress-time stablecoin substitution.

3.2 Cross-exchange spatial arbitrage

We compute log-price differentials for BTC/USDT (Binance vs. Kraken; Coinbase vs. Kraken) and BTC/USD (Coinbase vs. Kraken). An important caveat applies to the USDT cross-exchange channel: it captures dispersion in the USDT numeraire, so a USD-based arbitrage would require an additional USDT/USD conversion leg unless USDT/USD quotes are identical across venues. For Binance–Kraken BTC/USDT, the full-sample mean is near zero ($+0.007$ bps), while calm-period means remain economically small (pre-SVB -0.08 bps, post-SVB $+0.39$ bps; combined calm mean $\approx +0.16$ bps), consistent with tight cross-venue integration in this liquid channel. Coinbase–Kraken BTC/USD shows a persistent positive full-sample mean ($+2.18$ bps), consistent with a structural premium on the primary U.S. fiat gateway.

Figure 2: Cross-exchange basis: BTC/USDT (Binance vs. Kraken, Coinbase vs. Kraken) and BTC/USD (Coinbase vs. Kraken). The fiat channel (Panel C) shows a persistent positive basis that remains elevated post-crisis.

During the crisis, cross-exchange differentials widen alongside the intra-exchange deviations documented above (Figure 2): crisis-period mean absolute cross-exchange basis rises relative to calm periods in all channels (Table 10). This joint widening of intra- and cross-exchange dislocations is consistent with broad market stress rather than a single-venue disruption. Notably, the Coinbase–Kraken fiat premium did not fully normalize after the crisis, suggesting it may partly reflect structural microstructure differences that stress amplified rather than created.

3.3 Liquidity, volatility, and volume dynamics

We quantify liquidity across quote currencies using two standard microstructure estimators applied to the same 1-minute data. The Roll (1984) effective spread uses the daily serial covariance of log returns: $\hat{s}_t = 2\sqrt{-\mathrm{Cov}(\Delta p_t, \Delta p_{t-1})}$, expressed in basis points [10]. The Amihud (2002) illiquidity ratio $\mathrm{ILLIQ}_t = |r_t|/\mathrm{DollarVol}_t$ measures price impact per dollar traded [1]. Because the public candle APIs do not provide historical level-2 order-book snapshots, we cannot observe quoted depth directly; we therefore use Amihud and daily traded dollar volume as complementary depth proxies.

Table 5 and Figure 3 report regime-level means. The Roll spread for Kraken BTC/USDC increases from 1.10 bps pre-crisis to 25.64 bps during the SVB window, while Kraken BTC/USDT holds near 3 bps throughout. Because the Roll estimator requires negative serial covariance (bid-ask bounce) and yields NaN

on days with non-negative covariance, regime means for thin pairs rest on few valid days (as few as 1–2 per regime for the 3-day crisis window); these Roll estimates are therefore indicative of direction rather than precise level differences. For Kraken BTC/USD during the crisis, covariance turns non-negative on all days, consistent with sustained price-momentum as BTC recovered from the March 10 selloff. The Amihud ILLIQ confirms that Coinbase BTC/USD is the deepest market across all regimes (ILLIQ < 0.01), while Kraken BTC/USDC is the least liquid by this metric across regimes (pre 31.97, crisis 13.28, post 23.53 in $\times 10^{-6}$ units). The Roll and Amihud measures therefore move differently in crisis: Roll indicates sharply wider effective spread, while Amihud declines from its pre-crisis level as crisis-time dollar volume surges.

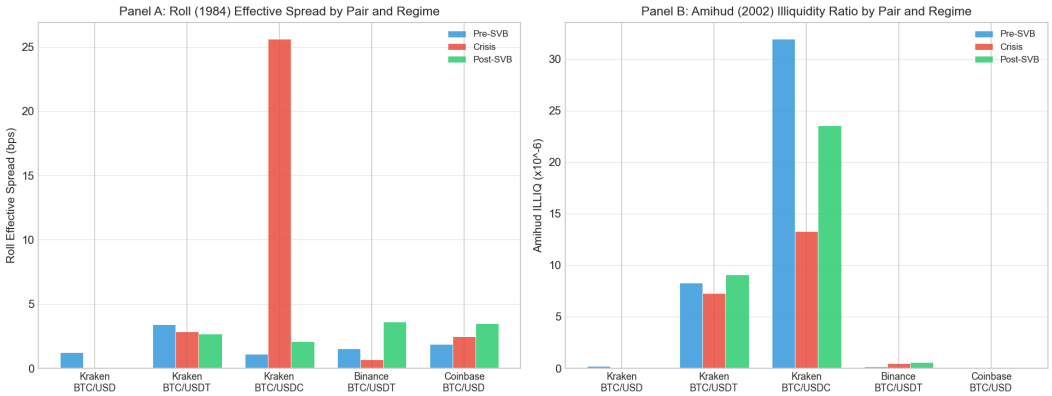

Figure 3: Panel A: Roll (1984) effective spread by pair and regime (bps). Panel B: Amihud (2002) ILLIQ ratio ($\times 10^{-6}$). Roll spread is undefined for days with non-negative serial covariance; regime means for thin pairs rest on few valid days (as few as 1–2 in the 3-day crisis window) and are therefore indicative. Roll is undefined for Kraken BTC/USD in stress periods (momentum regime).

Table 5: Roll (1984) effective spread (bps) and Amihud (2002) illiquidity ratio ($\times 10^{-6}$) by pair and regime. Roll spread estimated from daily serial covariance of 1-minute log returns; NaN days (non-negative covariance) excluded from means. N is the number of valid Roll days per regime; some regime means rest on few days and are indicative. $\text{ILLIQ}_t = |r_t|/\text{DollarVol}_t$, daily average.

Pair	Roll_{Pre}	$\text{Roll}_{\text{Crisis}}$	$\text{Roll}_{\text{Post}}$	N_{Pre}	N_{Crisis}	N_{Post}	$\text{ILLIQ}_{\text{Pre}}$	$\text{ILLIQ}_{\text{Crisis}}$	$\text{ILLIQ}_{\text{Post}}$
Kraken BTC/USD	1.260	—	—	4	0	0	0.232	0.050	0.068
Kraken BTC/USDT	3.390	2.850	2.670	1	1	1	8.299	7.266	9.074
Kraken BTC/USDC	1.100	25.640	2.070	2	2	2	31.970	13.282	23.530
Binance BTC/USDT	1.560	0.670	3.630	6	1	5	0.163	0.461	0.591
Coinbase BTC/USD	1.880	2.490	3.490	5	1	1	0.005	0.003	0.003

HAC regressions (Newey–West, 60 lags) of B_t on a crisis dummy, realized volatility, and intra-minute range (Table 6) show a large and significant USDC crisis coefficient ($+4.30$ bps, $p < 0.001$) [9]. Explanatory power remains modest ($R^2 = 0.041$ for USDC; 0.012 for USDT), so we interpret these coefficients as partial associations consistent with—not sufficient to fully explain—the liquidity evidence.

Daily dollar-volume depth proxies point in the same direction: Coinbase BTC/USD is deepest (median volume about \$239M/day pre-SVB, \$407M/day in crisis, \$664M/day post-SVB), while Kraken BTC/USDT and Kraken BTC/USDC are materially thinner (roughly \$6–17M/day and \$3–16M/day across regimes).

Table 6: HAC Regressions of Adjusted Residual B_t on Crisis Dummy, Realized Volatility, and Range Proxy (Kraken, Newey–West 60 lags)

	USDC Channel		USDT Channel	
	Coef.	p-value	Coef.	p-value
Constant	-0.079	0.833	-0.656	< 0.001
Crisis	$+4.295$	< 0.001	$+1.655$	< 0.001
RealizedVol (60m)	$+0.004$	0.928	-0.029	0.179
Range Proxy	$+0.048$	0.005	-0.013	0.082
R^2	0.041		0.012	
N	26,489		26,489	

OLS with HAC standard errors (Newey–West, 60 lags). Dependent variable: B_t (bps).

Volume dynamics tell a complementary story, but they are venue-specific. During the crisis, aggregate quote-currency shares are USD 80.3%, USDT 15.4%, and USDC 4.3% (vs. pre-SVB USD 81.3%, USDT 16.6%, USDC 2.1%), showing that USDC share rises in the acute window because activity concentrates in Kraken BTC/USDC while Binance BTC/USDC share falls. Post-crisis, share rotates back toward USD dominance (USD 86.9%, USDC 1.0%). This pattern indicates fragmentation and venue reallocation during stress rather than a uniform migration away from all USDC-quoted markets.

Realized volatility (60-minute rolling standard deviation of 1-minute simple returns, scaled to bps/hr via $\sqrt{60}$) reveals pronounced asymmetry across quote currencies. Kraken BTC/USDC peaks at $\approx$650 bps/hr on March 11, while crisis-average realized volatility is about 137 bps/hr for BTC/USDC versus 63–67 bps/hr for BTC/USD and BTC/USDT (roughly 2.1–2.2$\times$ higher in baseline all-sample windows). Under strict no-FF filtering, 60-minute crisis rolling windows retain only a small fraction of USDC/USDT observations (Table 2), so we treat exact volatility ratios as indicative rather than structurally identified constants. This excess volatility is consistent with compounding of asset risk and numeraire risk in BTC/USDC during stress. Post-crisis, USDC volatility normalizes but remains slightly elevated.

The distributional impact is visible in adjusted-residual diagnostics from the generated outputs. During the crisis, the USDC B_t distribution shifts rightward (mean $+5.3$ bps) and develops heavy tails extending beyond ± 100 bps, compared to a pre-SVB distribution concentrated within ± 10 bps; USDT tails widen more modestly. For risk management, this implies that positions denominated in stablecoins carry latent peg-break tail risk that is hard to detect in calm periods—standard VaR or margin models calibrated only on pre-crisis data can materially understate stress-period exposure. This tail channel is directionally aligned with the reserve-credibility mechanisms targeted by GENIUS Act provisions.

Distributional robustness. A Chow (1960) structural-break test at March 10 00:00 UTC strongly rejects parameter stability for both the unadjusted USDC dispersion ($F = 2{,}089$, $p < 10^{-16}$) and the adjusted residual ($F = 230$, $p < 10^{-16}$), formally validating the exogenous regime split [2]. Extending the test to the AR(1) dynamics of B_t ($X_t = c + \rho X_{t-1} + \varepsilon_t$) also rejects ($F = 71$, $p < 10^{-16}$): mean-reversion speed itself shifts at the break ($\hat{\rho}_{\text{pre}} = 0.49$, $\hat{\rho}_{\text{post}} = 0.37$), consistent with faster arbitrage correction under active crisis trading. Table 7 reports higher-moment diagnostics. USDC crisis B_t exhibits excess kurtosis of 12.9 (Fisher) and positive skewness of $+1.01$, versus pre-SVB excess kurtosis of 3.8 and near-zero skewness; USDT crisis kurtosis is much lower (1.4), confirming that the tail blowout is USDC-specific. Jarque–Bera tests reject normality for all B_t channels and regimes (JB > 300, $p \approx 0$ throughout), with USDC crisis the most extreme (JB $= 30{,}498$), formally justifying the non-Gaussian caveat in Section 2.3. The cross-stablecoin Pearson correlation between USDC and USDT adjusted residuals drops from $\rho = 0.44$ pre-SVB to $\rho = 0.16$ during the crisis, recovering to $\rho = 0.38$ post-SVB. This decorrelation during stress is consistent with divergent dynamics—USDC absorbing reserve-credibility risk while USDT serves as a short-term safe haven—rather than a common market-wide shock affecting both stablecoins symmetrically.

Table 7: Higher-moment diagnostics for adjusted residual B_t by regime. Excess kurtosis (Fisher) measures tail heaviness beyond Gaussian ($= 0$). Min/Max are in basis points.

Channel	Regime	Skewness	Ex. Kurt.	Min	Max
USDC	Pre-SVB	0.00	3.8	-36.9	32.0
USDC	Crisis	1.01	12.9	-165.2	180.6
USDC	Post-SVB	0.30	2.8	-43.2	79.4
USDT	Pre-SVB	-0.36	5.2	-36.4	34.5
USDT	Crisis	-0.11	1.4	-26.8	29.9
USDT	Post-SVB	-0.03	2.0	-37.8	36.2

3.4 Price discovery

Table 8: Johansen Cointegration and VECM Price Discovery (Primary Kraken Channels, No-FF Sample)

Channel	Rank	k_Δ	Trace$_{r=0}$	α_{USD}	α_{other}	IS$_{\text{USD}}$ [0.69, 0.77]	Leader
BTC/USD vs BTC/USDC	0	2	7.71	—	—	—	undetermined
BTC/USD vs BTC/USDT	1	8	27.03	0.0079	0.0118	0.73	BTC/USD

95% critical value for trace $r = 0$: 15.49. IS = Hasbrouck (1995) midpoint; bounds [0.69, 0.77].

Table 8 shows that Kraken BTC/USD vs. BTC/USDT rejects no cointegration (rank $= 1$; stable at $k_\Delta \pm 1$). We compute Hasbrouck (1995) Information Share (IS) bounds via both Cholesky orderings of the VECM residual covariance Σ_u [7]: IS$_{\text{USD}} \in [0.69,\ 0.77]$, midpoint **0.73**. For this Kraken USD–USDT channel and sample window, BTC/USD accounts for approximately 73% of permanent price discovery. The bounds are tight (width 0.08), so leadership is not driven by Cholesky ordering within this channel. This is consistent with the fiat anchor being deeper and more informationally efficient: the residual covariance Σ_u shows $\sigma_{\text{USD,USDT}}/\sqrt{\sigma_{\text{USD}}\sigma_{\text{USDT}}} = 0.889$, confirming very high co-movement. For BTC/USD vs. BTC/USDC, the no-FF specification does not reject rank $= 0$, consistent with instability of the long-run relation over this short event window.

BH/FDR-corrected Granger tests (Table 9) show directional asymmetry in the USDC channel: BTC/USD $\rightarrow$ BTC/USDC is significant, while the reverse direction is not significant after FDR correction. This pattern is consistent with fiat-to-stablecoin price adjustment. The USDT and cross-exchange channels show bidirectional significance, consistent with active two-way arbitrage in those more liquid and better-functioning channels.

Table 9: Granger Causality Tests with BH/FDR Correction (Secondary Evidence)

Channel	F-stat	p-value	Sig($p < 0.05$)	q-value (BH/FDR)	Sig($q < 0.0$
BTC/USDC → BTC/USD (Kraken)	1.512000	0.127898	No	0.127898	No
BTC/USD → BTC/USDC (Kraken)	332.178000	0.000000	Yes	0.000000	Yes
BTC/USDT → BTC/USD (Kraken)	3.825000	0.000034	Yes	0.000046	Yes
BTC/USD → BTC/USDT (Kraken)	385.271000	0.000000	Yes	0.000000	Yes
Kraken USDT → Binance USDT	7.265000	0.000000	Yes	0.000000	Yes
Binance USDT → Kraken USDT	476.065000	0.000000	Yes	0.000000	Yes
Kraken USD → Coinbase USD	3.709000	0.000055	Yes	0.000063	Yes
Coinbase USD → Kraken USD	130.133000	0.000000	Yes	0.000000	Yes

3.5 Arbitrage after fees

We evaluate arbitrage using explicit trade construction: 3-leg triangular for intra-exchange stablecoin channels, 2-leg pre-funded buy/sell for cross-exchange. Two cost bounds apply:

$$\text{cost}_t^{\text{fee}} = n_{\text{legs}} \cdot f, \tag{6}$$

$$\text{cost}_t^{\text{fee+slip}} = n_{\text{legs}} \cdot f + \sum_{\ell=1}^{n_{\text{legs}}} \tfrac{1}{2}\,\text{RangeProxy}_{\ell,t}, \tag{7}$$

with $f = 5$ bps per taker leg.

Table 10: Arbitrage Profitability by Channel and Regime (5 bps/leg; 3-leg intra-exchange, 2-leg cross-exchange)

Channel	Regime	Cost Variant	%Profitable	AvgNetUncond (bps)
USDC/USD (Kraken)	Crisis	Fee-only	29.16	4.83
USDC/USD (Kraken)	Crisis	Fee+slip	6.72	0.64
USDC/USD (Kraken)	Post-SVB	Fee-only	9.38	0.56
USDC/USD (Kraken)	Post-SVB	Fee+slip	1.78	0.08
USDT/USD (Kraken)	Crisis	Fee-only	3.76	0.14
USDT/USD (Kraken)	Crisis	Fee+slip	0.37	0.01
Cross BTC/USD (CB–KR)	Crisis	Fee-only	11.32	0.72
Cross BTC/USD (CB–KR)	Crisis	Fee+slip	2.29	0.15
Cross BTC/USDT (BN–KR)	Crisis	Fee-only	11.85	0.49
Cross BTC/USDT (BN–KR)	Crisis	Fee+slip	1.81	0.05

The fee-only specification is an optimistic upper bound. Adding range-based slippage materially reduces both frequency and net capture: in the intra-exchange USDC channel, crisis profitability falls from 29.16% to 6.72% (unconditional net 4.83 to 0.64 bps); in the cross-exchange Coinbase–Kraken BTC/USD channel, crisis profitability falls from 11.32% to 2.29%. Cross-exchange results assume pre-funded inventory and abstract from transfer delays and exchange-specific fee tiers. No-FF sensitivity (Table 2) preserves the same qualitative ranking (fee+slippage $\ll$ fee-only), with moderate level shifts in USDC channels on thinner samples. Fee sensitivity confirms the qualitative pattern: at $f = 3$ bps (maker-taker rebate scenario), USDC crisis fee+slippage profitability rises to 14.0%; at $f = 10$ bps (retail tier), it falls to 1.0%. The narrow post-cost net opportunities, even at peak stress, are consistent with limited executable arbitrage once realistic frictions are imposed.

4 Regulatory Interpretation

The SVB episode suggests that the cross-currency fragmentation we document was driven primarily by banking-layer reserve-access risk. Circle held \$3.3B of its ≈\$40B USDC reserves (≈8%) at SVB [3]. When SVB failed on Friday evening March 10, those reserves became inaccessible with no guaranteed redemption timeline, coinciding with the de-peg (Kraken trough \$0.874) and the $+320$ bps dispersion blowout. The two-layer half-life contrast sharpens this diagnosis: during the crisis, the adjusted basis B_t (cross-market arbitrage residual) mean-reverted with a half-life of ≈ 0.6 minutes, while the USDC peg deviation itself had a half-life of ≈ 572 minutes ($\hat{\rho} = 0.999$, ADF $p = 0.44$). This $\sim 940\times$ gap localizes the persistent dislocation to the peg layer—precisely the layer where reserve-access risk operates—and not to a breakdown in cross-market arbitrage efficiency. Near-par re-anchoring was rapid after the trough (first print ≥ 0.999 on March 13 UTC), while sustained recovery (daily minimum ≥ 0.999) was not reached until March 16—five days after the trough—and only after emergency deposit guarantees [5].

The GENIUS Act (signed July 2025) targets four dimensions of this failure chain; we map its provisions to our results as hypothesis-consistency checks, not identified causal effects [11]. (i) *Reserve composition* mandates one-to-one backing in Treasuries, FDIC-insured deposits, and central bank reserves. Table 11 maps assumed 25%–75% mitigation of reserve lock-up to implied crisis D_t of roughly 240, 160, and 80 bps; this linear mapping is illustrative, as actual de-peg dynamics are likely nonlinear. (ii) *Timely-redemption requirements* target panic amplification: during the SVB weekend, holders could not redeem for five days, turning a reserve-access problem into the pricing dislocation we document. The peg-layer half-life of 572 minutes—versus 1.06 minutes pre-SVB for the same series ($\hat{\rho}_{\text{pre}} = 0.521$)—quantifies the duration of this redemption-driven freeze. (iii) *Bankruptcy super-priority* targets the extreme B_t tail risk: crisis P99 reaches 81.4 bps versus pre-crisis P99 of 13.6 bps (a $6\times$ tail expansion), with excess kurtosis 12.9 and tails beyond ± 100 bps. (iv) *Transparency* (monthly reserve disclosure with CEO/CFO certification) closes the information channel: traders did not know USDC's SVB exposure until Circle's Friday-night announcement, consistent with the sharp USDC–USDT divergence ($D_t = +320$ vs. $+58$ bps premium) and the USDC–USDT decorrelation ($\rho = 0.16$) we document.

Table 11: GENIUS Act scenario range (illustrative, not structural causal identification). The table maps assumed mitigation of reserve lock-up shock into implied crisis D_t means, anchored to observed pre- and crisis-period moments. Tail benchmark for context: USDC B_t crisis P99 = 81.4 bps versus pre-crisis P99 = 13.6 bps.

Scenario	Assumed mitigation of lock-up shock (%)	Implied crisis D_t mean (bps)	Reduction vs observed (bps)	Policy mapping
Observed baseline	0	320.0	0.0	SVB/No GENIUS baseline
Low mitigation	25	240.0	80.0	Reserve composition + redemption + transparency
Moderate mitigation	50	160.0	159.9	Reserve composition + redemption + transparency
High mitigation	75	80.1	239.9	Reserve composition + redemption + transparency
Full-mitigation lower bound	100	0.1	319.9	Reserve composition + redemption + transparency

The persistence of cross-exchange fiat deviations post-crisis (Coinbase–Kraken BTC/USD $+2.18$ bps) and the $26\times$ depth gap between Coinbase BTC/USD ($407M/day crisis median) and Kraken BTC/USDC ($16M/day; Table 12) show that reserve-layer reforms alone may not eliminate spatial fragmentation; additional market-structure measures—cross-venue price feeds, unified clearing—may also be needed.

Table 12: Daily traded dollar volume depth proxy by pair and regime (median, USD millions/day). Because historical L2 order-book snapshots are unavailable in the candle APIs used here, dollar volume and Amihud ILLIQ are used as complementary depth proxies.

Pair	Median DollarVol Pre-SVB (MM/day)	Median DollarVol Crisis (MM/day)	Median DollarVol Post-SVB (MM/day)
Kraken BTC/USD	59.50	160.36	203.77
Kraken BTC/USDT	5.55	13.84	17.37
Kraken BTC/USDC	2.56	15.65	5.52
Binance BTC/USDT	52.44	73.92	71.46
Coinbase BTC/USD	239.07	407.27	664.46

5 Conclusion

This paper decomposes the cross-currency pricing fragmentation triggered by the SVB collapse into a large stablecoin-driven component (USDC crisis-mean $D_t \approx +320$ bps) and a smaller adjusted residual ($B_t \approx +5.3$ bps) that mean-reverts with half-lives of 0.54–1.10 minutes across robustness specifications. Crucially, the decomposition reveals a two-layer dynamic: the adjusted basis B_t mean-reverts in ≈ 0.6 minutes during crisis, while the USDC peg deviation half-life extends to ≈ 572 minutes—a $\sim 940\times$ gap that localizes persistent dislocation to the peg (reserve-access) layer, not to a breakdown in cross-market arbitrage. A Chow structural-break test formally validates the regime split at SVB onset for both dispersion levels ($F = 2{,}089$) and AR(1) dynamics ($F = 71$, both $p < 10^{-16}$), confirming that both the magnitude and speed of mean reversion shift at the break. The crisis is USDC-specific in its distributional signature: excess kurtosis of 12.9 (vs. 3.8 pre-SVB and 1.4 for USDT), with P99 expanding from 13.6 to 81.4 bps ($6\times$ tail blowout), and the cross-stablecoin adjusted-residual correlation drops from $\rho = 0.44$ to $\rho = 0.16$, consistent with divergent reserve-credibility dynamics rather than a symmetric common shock. The simultaneous USDT premium ($+58$ bps), venue-specific volume-share reallocation (USDC pair share spiking from 2.1% to 4.3% in crisis before collapsing to 1.0% post-crisis), and USDC-specific volatility amplification (2.1–$2.2\times$, with realized volatility remaining elevated post-crisis at 92 bps/hr versus 78 bps/hr for BTC/USD) reinforce the same conclusion: stress-time fragmentation is stablecoin-specific and reserve-driven.

Fiat-quoted BTC leads price discovery over USDT-quoted BTC on Kraken (Hasbrouck IS midpoint 0.73, bounds $[0.69, 0.77]$), while the USDC channel does not sustain a stable cointegrating relationship over this window. Post-cost arbitrage opportunities narrow sharply under realistic frictions: intra-exchange USDC crisis profitability falls from 29% to 7% (fee+slippage at $f = 5$ bps)—versus 0% pre-SVB under the same cost specification—and cross-exchange BTC/USD falls to 2.3%, with fee sensitivity confirming this pattern across $f \in \{3, 5, 10\}$ bps. The $26\times$ depth gap between the deepest fiat market (Coinbase BTC/USD, $407M/day crisis median) and the thinnest stablecoin channel (Kraken BTC/USDC, $16M/day) explains why even peak-stress opportunities are largely unexecutable at scale.

These findings support a directional policy implication: the GENIUS Act's reserve-composition, timely-redemption, transparency, and bankruptcy super-priority provisions target precisely the failure chain the

SVB episode exposed—reserve inaccessibility, information asymmetry, panic amplification, and peg instability [11]. The two-layer half-life result sharpens this mapping: because the persistent crisis dislocation resides in the peg layer (572-minute half-life) rather than the arbitrage layer (0.6-minute half-life), reforms that reduce reserve-access risk—the peg layer—should compress D_t more effectively than measures targeting cross-market efficiency. An illustrative counterfactual (Table 11) maps 50% mitigation of the reserve lock-up to implied crisis $D_t \approx 160$ bps, though actual dynamics are likely nonlinear. A natural next step is a post-GENIUS replication study during the next stress event, testing whether the Act's provisions are associated with smaller D_t and B_t tails, weaker stress-time stablecoin substitution, and narrower arbitrage windows relative to the SVB baseline. The persistence of cross-exchange fiat deviations post-crisis and the depth asymmetry documented above suggest that reserve-layer reforms alone may be insufficient, and that additional market-structure measures—cross-venue price feeds, interoperability mandates, unified clearing—may be needed to compress residual spatial fragmentation.

Interpretation Boundaries. This evidence is channel- and window-specific (March 1–21, 2023 UTC) and observational; policy passages are consistency mappings, not identified treatment effects. Four robustness checks anchor interpretation: (i) HAC intervals preserve the directional signs of headline means; (ii) strict no-FF filters preserve core rank-order conclusions while attenuating thin-channel levels; (iii) rank-stable cointegration is supported for Kraken BTC/USD vs. BTC/USDT but not BTC/USDC; and (iv) Chow tests confirm the regime split formally for both levels and dynamics. Metrics with low no-FF retention (notably 60-minute rolling volatility) are interpreted by direction and rank-order, not point precision; Roll regime means for thin pairs rest on few valid days and should be read as directional.

Reproducibility Note

All in-paper exhibits are deterministic outputs of `run_all.py` (via `src/03_analysis_and_figures.py` and `src/06_three_fixes.py`), with artifact-label integrity checks in `src/04_tex_integrity_check.py`. Provenance artifacts for prose-only headline values are generated in the same pipeline, including quote-currency volume-share summaries, realized-volatility headline snapshots, and peg-recovery threshold timestamps.

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